Ashwin Chembu

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PROFILE

Backend engineer building Python and Node.js APIs, storage integrations, authentication, and event-driven workflows. Focuses on secure service boundaries, durable data handoffs, and repeatable validation.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Services / APIs: FastAPI, Flask, Node.js, Express, REST APIs, OpenAPI, MCP

Storage / Cloud: MongoDB, Redis, AWS S3, Firebase, Render

Reliability: HMAC webhooks, authentication, validation, rate limiting, pytest

EXPERIENCE

Business Technology Solutions Associate | ZS Associates **South San Francisco, CA | Jun 2024-Present**

- Built **Python**/Flask endpoints connecting AWS S3 and Smartsheet APIs to automate document storage and cloud handoffs.
- Built React/FastAPI RAG assistants with AWS document ingestion and Redis services for enterprise knowledge search.
- Built Snowflake SQL pipelines over **327K+** records for identifier matching, address comparison, and quality reporting.
- Contributed FastAPI/Uvicorn model workflows and pytest checks to validate enterprise RAG services and AI behavior.

Founder and Lead Developer | Tallyrus **Jun 2023-Sep 2025**

- Built Flask services with **MongoDB** persistence and S3 storage to manage documents and support essay-feedback flows.
- Integrated React/TypeScript and **Express** flows with Google OAuth, **MongoDB,** sessions, and protected product routes.
- Built **Python**/LangChain essay-feedback workflows with Pinecone and OpenAI for retrieval-augmented educator review.

Data Analyst / Web Development Lead | Tech4Good at UCSC **Mar 2023-Jun 2025**

- Implemented 20+ attributed Node/**Express** routes with **MongoDB** persistence and logging for chat, text entry, and polls.
- Separated chat, text, and poll state in three **MongoDB** databases, using change streams to synchronize edits and votes.
- Built TypeScript/Figma API chat, polling, and export workflows in a collaborator-owned repository for classroom collaboration.

PROJECTS

Private Filedrop | Secure File Exchange **Jun 2026-Aug 2026**

- Built password-protected uploads and API-key-protected **Express** APIs with **OpenAPI** docs for secure file exchange.
- Added local/S3-compatible storage, migration tooling, version history, and unguessable links for durable file handoffs.
- Exposed **MCP** file tools with HMAC-SHA256 webhooks and timing-safe credential checks for signed service events.
- Deployed **Express** on Render and documented browser, REST, **OpenAPI,** and **MCP** workflows for cross-session handoffs.

Forge | Personal Fitness Intelligence **Personal project**

- Integrated Hevy/Oura APIs and Apple Health webhooks with FastAPI/**MongoDB** to unify training, nutrition, and recovery.
- Implemented configurable Hevy/Oura synchronization, on-demand refresh, and CSV ingestion to keep fitness records current.
- Compared performance with a 12-week hypertrophy plan to score workout adherence, nutrition compliance, and recovery.

Lume | AI Meeting Intelligence **Dec 2025-Jan 2026**

- Built React and Node/**Express** flows with **MongoDB**/S3 to capture, store, search, and query meetings and project files.
- Designed RAG with vector retrieval, intent routing, evidence extraction, and generator/critic passes for cited answers.
- Built reusable Project Brain knowledge bases retaining transcript/file terminology and relationships for grounded Q&A.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu